



UNIT – II

MANUFACTURING PROCESS

Manufacturing is an activity of turning raw Material into finished Products to be used for some Purpose. It also involves assembly of Parts to make final Product.

Types of Manufacturing Processes

1) Primary manufacturing Processes: It involves the Conversion of initial Material into Semi-finished Products

Ex: Casting, rolling, forging, and extrusion. etc.

2) Secondary Manufacturing Processes: In this Process the accurate size of final Products are obtained.

Ex: Coining, drawing, sheet-forming etc.

Based on the Primary and Secondary Manufacturing process We Can classified as

(a) Casting [Starting Material is liquid]

(b) Metal -forming (or) Metal Working Process: [Starting Material is "Solid which consists of Plasticity.

(c) Machining (or) cutting: Starting Material is solid which extra material is removed.

(d) Powder metallurgy (or) Particulate process: starting material is powder which is compacted and sintered.

(e) Joining: In this process two or more components are joined together to produce the required product. Examples are welding, brazing, soldering etc.

(f) Assemble processes: the assemble processes for machines and mechanisms is the part of the manufacturing process concerned with the different joining of the finished parts.

(g) Surface treatment process: Unwanted material like oil, dirt, will, removed by this process.

Selection of manufacturing Process:

Many factors are involved in Selection of Manufacturing Process. Some of the considerations are.

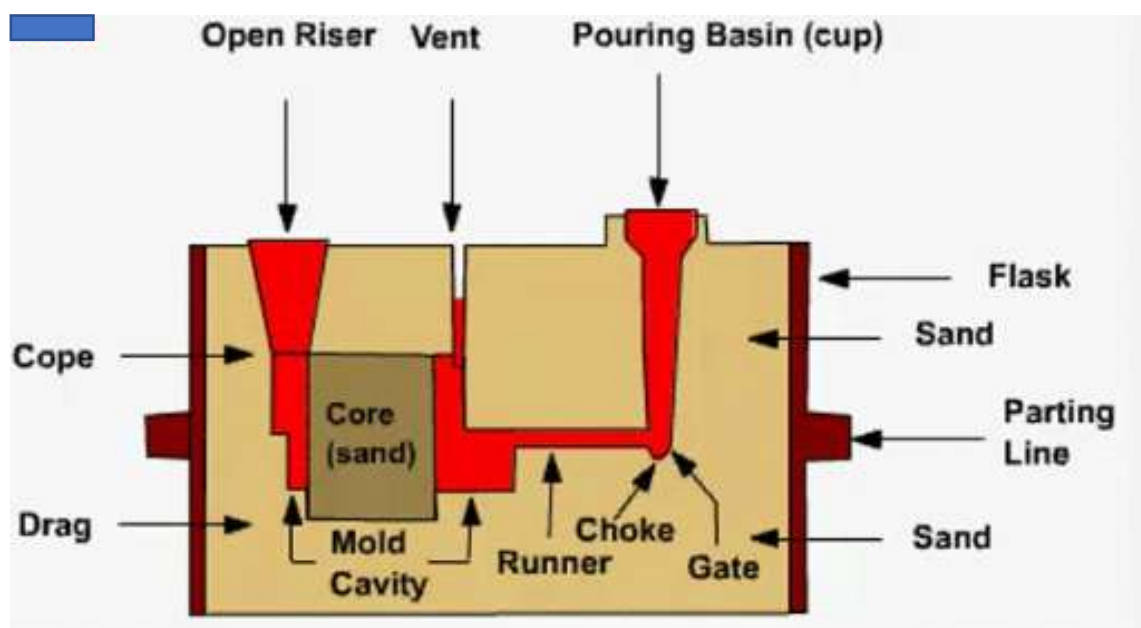
- 1) Material to be used
- 2) Shape and size of the Components
- 3) Accuracy and Surface -finish
- 4) volume of Production
- 5) Economy.

Importance

Now-a-days the economic development of Country is completely based on Material resources, skill and energies expand in utilising resources. ie. Manufacturing. Manufacturing is a Series of complex interaction between Materials, Machines, energy and People. It begins with Certain of individual Parts that will finally assembled to Produce final Products

Casting

Casting is defined as “the shaping of Material in liquid State”. Here the liquid Metal is poured in a mould Cavity, usually made of Sand, and it is allowed to Solidify after completion of Solidification the part is removed from mould either by breaking Core, Separating the two Parts of mould. The part obtained by casting is called "Casting (or) founding.



Drag: Lower-Moulding flask

Cope: Upper - Moulding flask

Pattern: It is replica of a final object-to be made. The mould cavity is made with help of Pattern.

Parting line: It is a dividing line b/w 2 Moulding flask.

Facing Sand: It is a small amount of Carbonaceous Material Sprinkled on the inner Surface of Moulding cavity to Provide better Surface finish.

Moulding Sand: It is a mixture of silica, clay, moisture with Certain Proportions.

Care: It is used for making hollow Cavities in Castings.

Pouring Basin: It is a Small-shaped Cavity at the top of the mould into which molten metal is Poured

Sprue: It is a Passage through the molten metal -from the Pouring basin.

Runner. It is a Passage ways, which molten metal flow is regulated before they reach the mould Cavity.

Gate: The actual entry Point through molten metal enters the mould cavity.

Riser: It is a reservoir of molten metal Provided in the Casting, So that hot metal Can flow back into the mould cavity.

Steps involved in Casting:

Steps involved in casting: Now-a-days 75% of the total Outlet of castings are made in Sand Moulding. So the steps involved in Saud Moulding are

1) Selection of mould box/ flask:

of Casting Based on the requirement of casting we can Select the mould, which must be of Proper size to adjust mould: Cavity, riser, and gating System etc.

2) Preparation of Drag:

- Initially the drag will be placed on bottom board
- "Facing Sand" is Sprinkled Care-fully around the drag Pattern' So that the Pattern does not Stick with moulding Sand during removal of Pattern
- The drag is filled with moulding Sand and rammed this sand around the Pattern, the ramming Process is close 3(or) 4 times.
- The excess amount of Sand is removed with help of Strike off bar.
- The drag is the rolled over by 180 and the Parting Sand is Sprinkled over on the top of drag.

3) Preparation of cope:

- The cope Pattern is placed on the help of dowel Pins.
- The pattern Sand is sprinkled carefully around the Cope Pattern.
- Runner Pin [Sprue] and Riser Pins are located in Vertical Position at Suitable locations. These are help to form Suitable Sizes of cavities for Pouring molten metal.
- Now the cope will be filled with moulding Sand and ramming Process is done. and excess Material is removed
- Remove sprue and riser Pins and Create vent holes. These are helps to "escape of gases during Pouring and Solidification of casting.

4) Cutting of Gate & Pouring of Metal:

- Remove both the Cope and drag Patterns, and repair the mould Suitable if needed and dressing is applied.
- The gate is then cut connecting the lower base of sprue basin with runner and then the mould Cavity.
- The Cope is then clamped with drag and the mould is ready for Pouring

Advantages:

1. Design flexibility
2. Reduced costs
3. Dimensional accuracy
4. Versatility in production

Disadvantages:

1. Lot of molten metal is wasted in riser & gating.
2. Casting may require machining to remove rough surfaces.

Applications:

Automobile parts (engine block, cylinder head)

Machine components (machine beds, pumps, valves)
Pipes and pipe fittings
Turbine blades (aerospace)
Railway wheels and parts
Manhole covers (construction)

FORMING

“Forming is a manufacturing process in which the shape of a solid metal is changed by plastic deformation under applied force, without removing or melting the material.”

Types of Forming Process:

1) Bulk forming:

Bulk forming is a forming process in which a large amount of metal is plastically deformed, usually under compressive forces, to change its shape.

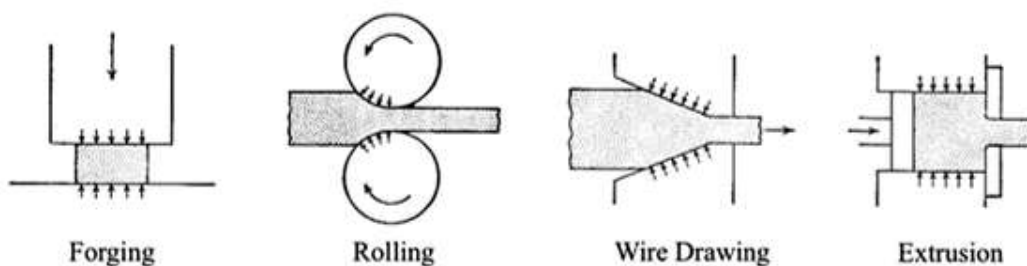
Examples:

Rolling: In this process, the work piece in the form of slab or plate is compressed between two rotating rolls in the thickness direction, so that the thickness is reduced. The rotating roll draws the slab into the gap and compresses it. The final product is in the form of sheet.

Forging: The work piece is compressed between two dies containing shaped contours. The die shapes are imparted into the final part.

Extrusion: In this, the work piece is compressed or pushed into the die opening to take the shape of the die hole as its cross section.

Wire drawing: similar to extrusion, except that the work piece is pulled through the die opening to take the cross-section.



2) Sheet metal Forming:

Sheet metal forming is a process in which thin metal sheets are plastically deformed to obtain the required shape without removing material.

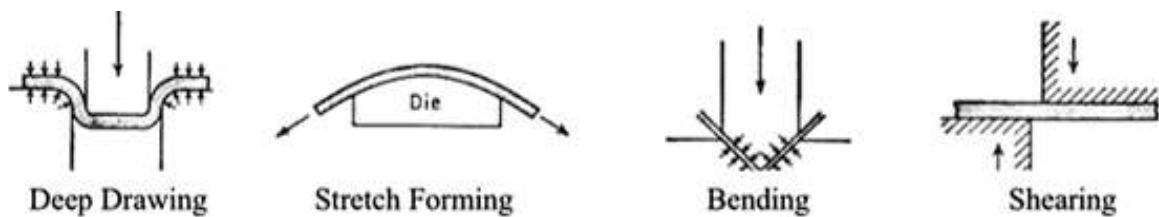
Examples:

Sheet forming: Sheet metal forming involves forming and cutting operations performed on metal sheets, strips, and coils. The surface area-to-volume ratio of the starting metal is relatively high. Tools include punch, die that are used to deform the sheets.

Bending: In this, the sheet material is strained by punch to give a bend shape (angle shape) usually in a straight axis.

Deep (or cup) drawing: In this operation, forming of a flat metal sheet into a hollow or concave shape like a cup, is performed by stretching the metal in some regions. A blank holder is used to clamp the blank on the die, while the punch pushes into the sheet metal. The sheet is drawn into the die hole taking the shape of the cavity.

Shearing: This is nothing but cutting of sheets by shearing action.



Advantages:

- Good surface finish
- High production rate (mass production possible)
- Less material waste
- High dimensional accuracy
- Suitable for complex shapes (thin components)
- Lightweight products

Disadvantages:

- Suitable only for thin sheets
- High initial tooling cost (dies, presses)
- Chances of defects (wrinkling, tearing)
- Limited thickness range
- Skilled operation required

Applications:

- Automobile body parts (doors, panels)
- Aircraft skin and structural parts
- Kitchen utensils (cups, containers)
- Electrical enclosures (switch boxes)
- Metal cans and packaging
- Roofing sheets and panels

Joining



Joining is one of the manufacturing processes that are used to assemble two or more solid components or structures together. It can be employed to permanently or temporarily join any type of engineering materials such as metals, ceramics, plastics and composites. In order to cater the need to join wide variety of materials in several ways, there exist various joining processes, which can be broadly classified into two groups

Temporary joining: As the name suggests, a temporary joint is not fixed. That means it allows easy and quick dismantling of jointed structures without breaking them. This is very useful for inspection, maintenance and repairing purposes. However, such joint may not necessarily be strong and leak-proof. Typical joining processes that allow temporary joints include fastening, cotter joint, knuckle joint, key joint, press fit, seam joint, etc.

2. Permanent joining: A permanent joint is meant to stay fixed for long duration. It does not allow disassembly of jointed structures without breaking them. It possesses challenge during inspection, maintenance and repairing purposes as sometimes component or structure parts are required to damage. However, it offers a sound, reliable and leak-proof joint with high load carrying capacity. Typical permanent joining processes include welding, riveting, soldering, brazing, coupling, adhesive joining, etc.

Advantages

1. Complex shapes possible
2. Large structures can be made
3. Different materials can be joined
4. Repair work easy

Disadvantages

1. Joint may be weak compared to base metal
2. Defects may occur
3. Inspection required
4. Some processes are costly

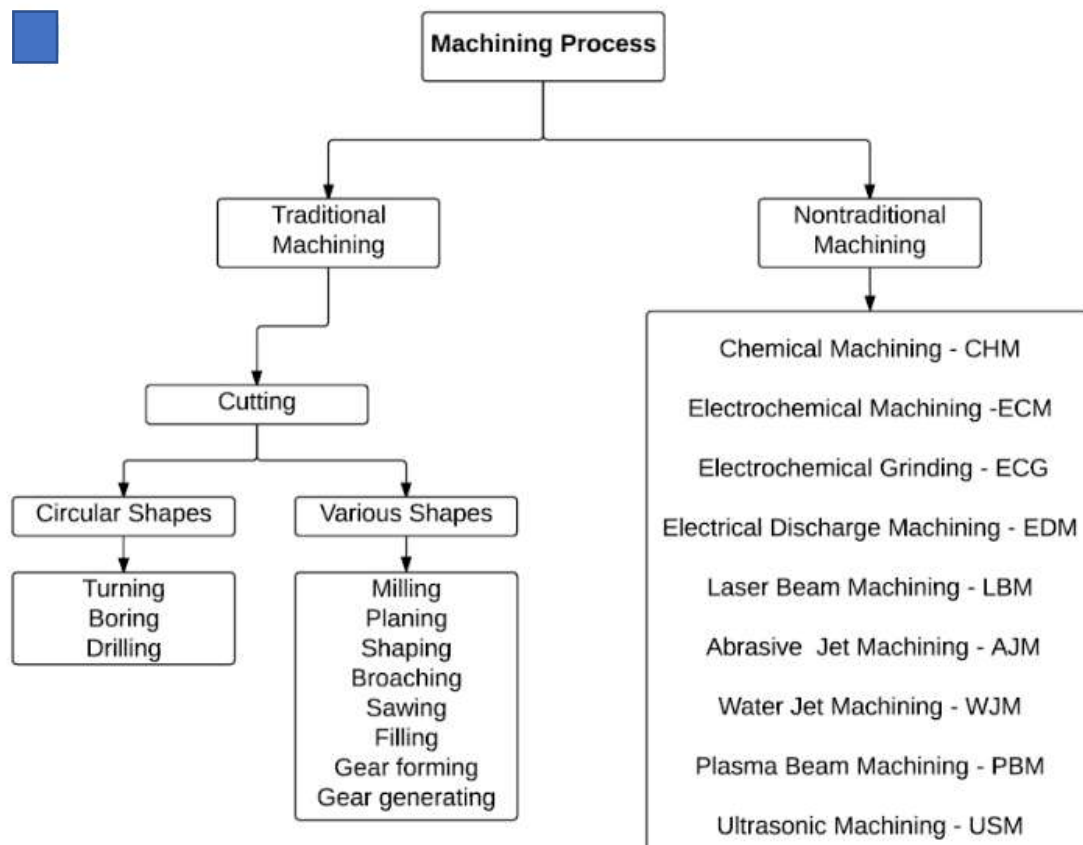
Applications

1. Automobile (car body, chassis)
2. Construction (bridges, buildings)
3. Aerospace (aircraft structures)
4. Electronics (circuit boards)
5. Shipbuilding

Machining

Machining is a manufacturing process in which material is removed from a workpiece in the form of chips using cutting tools to obtain the required shape, size, and surface finish.

Classification of Machining Process:



Traditional Machining process:

Traditional machining is a process where sharp cutting tools physically cut the material to get the required shape and size.

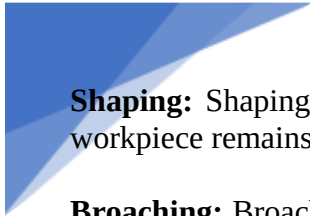
Turning: Turning is a machining process in which a rotating workpiece is cut by a single-point cutting tool to produce cylindrical shapes.

Boring: Boring is a machining process used to enlarge and finish an already existing hole using a single-point cutting tool.

Drilling: Drilling is a machining process used to create a circular hole in a workpiece using a rotating drill bit (multi-point tool).

Milling: Milling is a machining process in which material is removed from a workpiece using a rotating multi-point cutting tool to produce flat or complex surfaces.

Planing: Planing is a machining process in which the workpiece moves back and forth, and the cutting tool remains stationary to produce flat surfaces.



Shaping: Shaping is a machining process in which the cutting tool moves back and forth, while the workpiece remains stationary to produce flat surfaces.

Broaching: Broaching is a machining process in which material is removed using a multi-point cutting tool called a broach, where teeth are arranged in a series to remove material in one pass.

Sawing: Sawing is a machining process used to cut a workpiece into two or more parts using a toothed blade.

Filing: Filing is a machining/finishing process in which small amounts of material are removed using a file (multi-point cutting tool) to obtain smooth surface, correct size, and shape.

Gear generation: Gear generating is a machining process in which gear teeth are produced by a relative rolling motion between tool and workpiece, following the correct tooth profile.

Non traditional machining Process:

Non-traditional machining is a process in which material is removed without direct contact between tool and workpiece, using electrical, thermal, chemical, or high-energy methods.

Advantages:

- High accuracy and precision
- Good surface finish
- Suitable for complex shapes
- Can machine hard materials
- Tight tolerances can be achieved
- Good dimensional control
- Suitable for small batch production

Disadvantages:

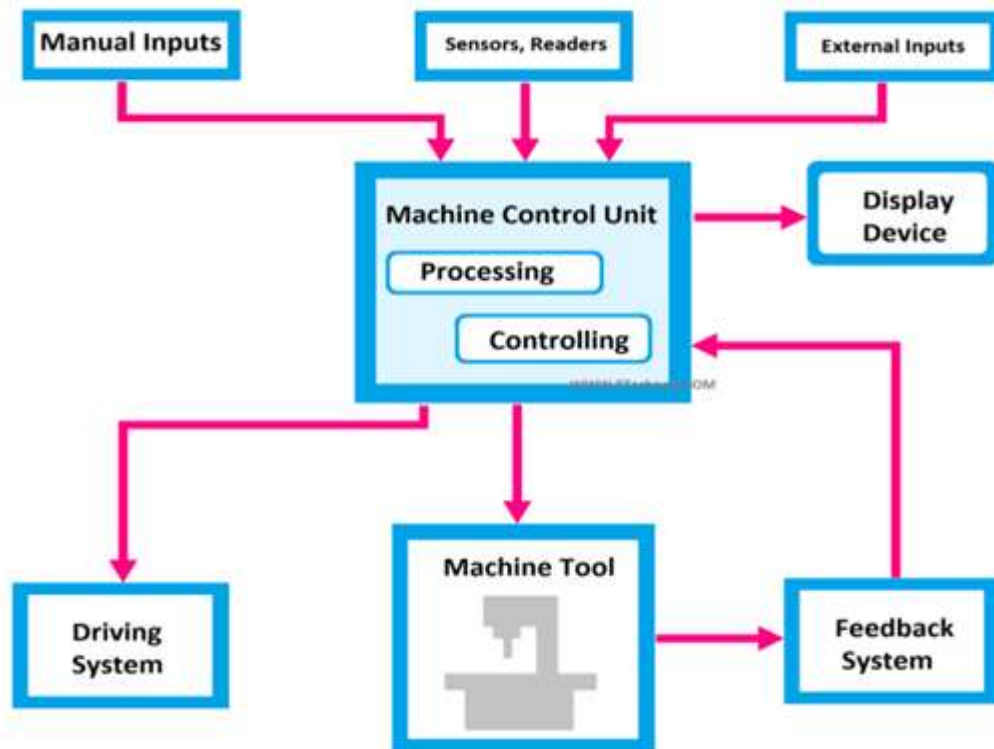
- High material wastage (chips)
- Time-consuming process
- High labor and machine cost
- Tool wear occurs
- Requires skilled operators
- Not economical for mass production
- Energy consumption is high

Applications:

- Manufacturing machine parts (shafts, gears, bolts)
- Automobile components
- Aerospace components
- Tool and die making
- Repair and maintenance works
- Precision components manufacturing

CNC MACHINES

Introduction to CNC machines: CNC means Computer Numerical Control. When computers are used to control a Numerical Control (NC) machine tool then the machine is called CNC machine. In other words the use of computers to control machine tools like lathe, mills, shaper etc is called CNC machine. The cutting operations performed by the CNC is called CNC machining, in CNC machining, programs are designed or prepared first and then it is fed to the CNC machine. According to the program, the CNC controls the motion and speed of the machine tools.



The main parts of the CNC machine are:

1) Input Devices: These are the devices which are used to input the part program in the CNC machine. There are three commonly used input devices and these are punching tape reader, magnetic tape reader and computer via RS-232-C communication.

2) Machine Control Unit (MCU): It is the heart of the CNC machine. It performs all the controlling action of the CNC machine, the various functions performed by the MCU are a. It reads the coded instructions fed into it. b. It decodes the coded instruction

3) Machine Tool: A CNC machine tool always has a slide table and a spindle to control the position and speed. The machine table is controlled in X and Y axis direction and the spindle is controlled in the Z axis direction.

4) Driving System: The driving system of a CNC machine consists of amplifier circuits, drive motors and ball lead screw. The MCU feeds the signals (i.e. of position and speed) of each axis to the amplifier circuits. And the actuated drive motors rotate the ball lead screw to position the machine table.

5) Feedback System: This system consists of transducers that acts like sensors. It is also called as measuring system. It contains position and speed transducers that continuously monitor the position and speed of the cutting tool located at any instant. The MCU receives the signals from these transducers and it uses the difference between the reference signals and feedback signals to generate the control signals for correcting the position and speed errors.

6) **Display Unit:** A monitor is used to display the programs, commands and other useful data of CNC machine



Different Types of CNC Machines:

There are several types of CNC machines

- 1) Mill CNC Machine
- 2) Lathe CNC Machine
- 3) Drilling CNC Machine
- 4) Grinding CNC Machine
- 5) Laser Cutting CNC Machine
- 6) Plasma Cutting CNC Machine
- 7) Water jet Cutting CNC Machine
- 8) EDM (Electric Discharge Machine)

Advantages:

- 1) It can produce jobs with the highest accuracy and precision than any other manual machine.
- 2) It can be run for 24 hours a day.
- 3) The parts produced by it have the same accuracy. There is no variation in the parts manufactured by a CNC machine.
- 4) A highly skilled operator is not required to operate a CNC machine. A semi-skilled operator can also operate accurately and more precisely.
- 5) Operators can easily make changes and improvements and reduces the delay time.
- 6) It has the capability to produce complex designs with high accuracy in minimum possible time.
- 7) The modern design software, allows the designer to simulate the manufacturer of his/her idea. And this removes the need for making a prototype or model and saves time and money.
- 8) Fewer workers are required to operate a CNC machine and save labor cost.

Disadvantages:

- 1) The cost of the CNC machine is very high as compared with a manually operated machine.
- 2) The parts of the CNC machines are expensive.
- 3) The maintenance cost in the case of CNC is quite high.
- 4) It does not eliminate the need for costly tools.

Application:

Almost every manufacturing industry uses CNC machines. With an increase in the competitive environment and demands, the demand for CNC usage has increased to a greater extent. The machine tools that come with the CNC are lathe, mills, shaper, welding, etc. The industries that are using CNC machines are the automotive industry, metal removing industries, industries of fabricating metals, electrical discharge machining industries, wood industries, etc.

3D printing:

3D printing is the process of creating a three dimensional object from a digital design by building up layers of material with a 3D printer. 3D printing is also known as additive manufacturing because material is added during the process of manufacturing to form the object. The opposite of this, known as subtractive manufacturing, is where material is cut away from a sheet or a block of material to form the object, for example manufacturing processes such as milling or laser cutting.

Where is 3D printing used: 3D printing is now being used across a wide range of industries including engineering, automotive, aerospace, construction, architecture, medicine, product design, food and fashion.

Examples:

- a. Used for rapid prototyping of parts and components in engineering, aerospace and automotive industry.
- b. Creation of custom jigs and templates for the automotive industry.
- c. In medicine, 3D printing from patient scans to assist in planning operations or for student training.
- d. Production of custom low cost prosthetics and disability aids.
- e. 3D printing shelters and humanitarian aids.
- f. Small enterprises creating personalised 3D printed products and gifts.
- g. 3D printing shoes,



Overview of the 3D printing process:

The 3D printing process consists of a number of stages from initial design to production of a 3D printed object.

1) 3D modeling: The 3D printing process starts with a 3D model. You can design your own 3D model using 3D CAD software such as TinkerCAD, you can also download free models that have been designed by others from 3D model repositories such as these can be edited in 3D CAD software if required to customise the designs to your own requirements.

2) Saving the 3D model: When the 3D model is complete, it needs to be saved as a suitable format. The most common 3D file type is .STL so save the file as a .STL file where possible. The Ultimaker software will also support the .OBJ .3MF formats, so if you cannot save as a .STL save as one of these.

3) Preparing the 3D model for printing: Before you can 3D print a CAD model, the model needs to be sliced and code needs to be generated that the machine can understand. This is done in special 3D printer slicing software. The Ultimaker slicing software is called Cura.

Advantages:

- Complex shapes can be easily created
- Less material waste
- Fast prototyping
- Easy customization
- No need for tooling
- Lightweight parts can be made
- Suitable for small production

Disadvantages

- Slow production speed
- Limited materials available
- High machine cost
- Poor surface finish (needs post-processing)
- Lower strength in some cases
- Size limitation
- Skilled operators required

Applications:

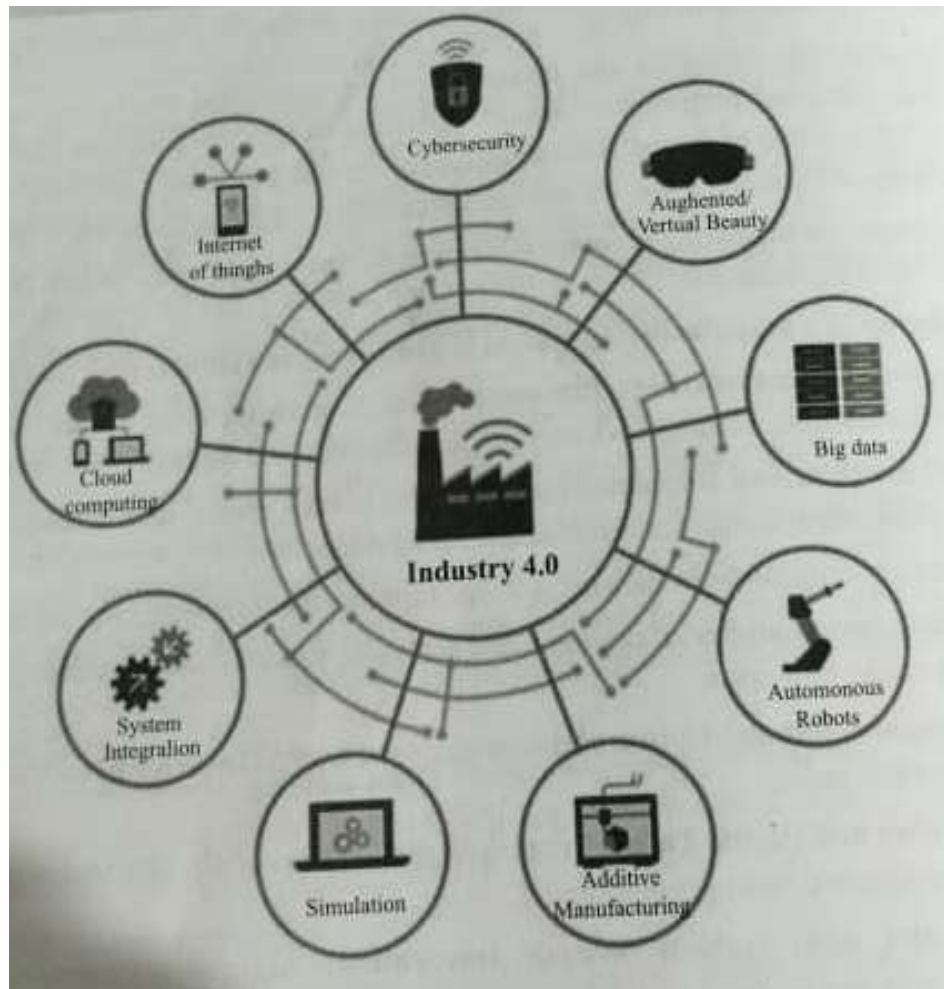
- Medical field (prosthetics, implants)
- Automobile industry (prototype parts)
- Aerospace (lightweight components)
- Education (models)
- Architecture (building models)
- Consumer products (toys, jewelry)
- Tooling (jigs and fixtures)

SMART MANUFACTURING

Smart Manufacturing is a modern approach to production that uses digital technologies to make manufacturing processes intelligent, automated, and efficient. It is a key part of **Industry 4.0**.

In smart manufacturing, machines, sensors, and systems are interconnected through the Internet of Things (IoT). These systems continuously collect real-time data, which is analyzed using advanced tools like Artificial Intelligence (AI) and Big Data Analytics. Based on this data, the system can automatically make decisions, optimize processes, and even predict machine failures.

The main technologies used in smart manufacturing include IoT, AI, cloud computing, robotics, and digital twin technology. These technologies help industries achieve automation, reduce human errors, and improve product quality. Smart manufacturing offers several advantages such as increased productivity, reduced waste, faster production, and predictive maintenance



Advantages of Smart Manufacturing:

- 1) Increased Efficiency
- 2) Increased Production
- 3) Improved Process Quality
- 4) Energy Savings
- 5) Reduced Fixed Costs
- 6) Increased Reactivity
- 7) Increased Competitiveness
- 8) Higher Productivity

Disadvantages of Smart Manufacturing:

- 1) High initial cost
- 2) Poorly designed systems
- 3) Lack of skilled workers
- 4) Organizational resistance

Applications:

- 1) Predictive maintenance
- 2) Data capture
- 3) Data analytics
- 4) Automation and machine building

Thermal Engineering

Thermal Engineering is a specialized branch of Mechanical Engineering that deals with the study, analysis, and application of heat energy and its conversion into mechanical work. It is based on the principles of Thermodynamics and Heat Transfer, which explain how heat is generated, transferred, and utilized in various systems.

Working Principle of Boilers:

A boiler is a closed pressure vessel used to generate steam by heating water with heat energy produced from fuel combustion. It works on the principle of Heat Transfer, where heat is transferred to water through conduction, convection, and radiation, causing it to convert into steam at required pressure and temperature. The generated steam is then used for various applications such as power generation, heating, and industrial processes, making boilers an essential component in thermal engineering systems.

Classification Of Boilers:

Boilers are classified based on different factors such as flow of fluids, position, pressure, fuel, etc.

1) Based on flow of water and Hot Gases:

- a) **Fire tube Boiler:** Hot gases are contained within tubes, with water surrounding them (e.g., vertical, Lancashire and locomotive boilers)
- b) **Water tube Boiler:** Water is contained within tubes, with hot gases surrounding them (e.g., Babcock and Wilcox, La-mount, Benson and Loeffler boilers).

2) Based on axis of Boiler

- a) **Horizontal:** Features a horizontal axis for the boiler shell (e.g., Lancashire and locomotive boilers).
- b) **vertical:** Possesses a vertical axis for the boiler (e.g., Cochran boiler)
- c) **Inclined:** Features an inclined axis

3) Based on Pressure

- a) **low :** Operates within the 3.5 to 10 bar pressure range (e.g., Cochran and Cornish boilers).
- b) **Medium:** Operates within the 10 to 25 bar pressure range (e.g., Lancashire and locomotive Boilers)
- c) **High:** Operates at over 25 bar pressure (e.g., Babcock & Wilcox boiler).

4) Based on Method of Firing

- a) **Internally Fired:** The furnace is situated within the boiler shell (e.g., Cochran and Lancashire boilers).

b) **Externally Fired:** Fuel combustion occurs outside the shell (e.g., Babcock & Wilcox)

5) **Based on circulation of boiler**

a) **Natural circulation:** Water circulation arises from density differences due to temperature variances (e.g., Lancashire, Babcock and Wilcox boilers).

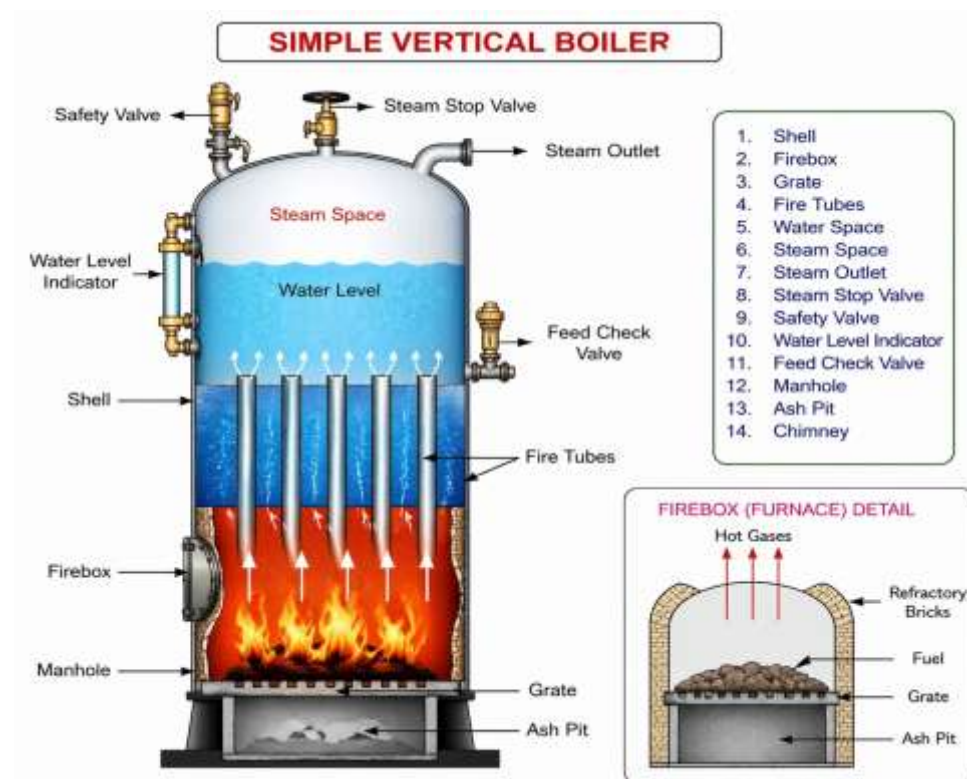
b) **Forced circulation:** Water circulation is achieved via a pump (e.g., Velox, La-Mount, Benson boilers).

6) **Based on Use**

a) **Stationary boilers:** Impractical for easy transportation (e.g., Lancashire and Babcock and Wilcox boilers)

b) **Portable boilers:** Designed for easy relocation (e.g., Locomotive boiler).

Simple Vertical Boiler:



Working:

- The fuel (coal) is fed into the grate through the fire hole and is burnt. The ash pit placed below the grate collect the ashes of the burning fuel.
- The combustion gas flows from the furnace, passes around the cross tubes and escapes to the atmosphere through the chimney.
- Water goes by natural circulation due to convection currents, from the lower end of the cross tube and comes out from the higher end.

Construction:

- Such boilers have steam generation capacity up to 1000 kg per hour

- Maximum steam pressure up to 10 bar.
- Size of the boiler ranges from 0.6 m diameter to 2 m diameter
- height from 1.2 m to 4 m high.
- The working pressure of the simple vertical boiler does not exceed 70 N/cm².
- Total heating surface area is about 7-10 times grate area.

Advantages:

- The initial cost is low because of fewer parts.
- Maintenance cost is low.
- Working is simple.
- It is easy to install and replace.
- It occupies a small space on the ground.
- These type of boilers have water level tolerance.

Disadvantages:

- The vertical design of this boiler limits its work in many places.
- Steam production is limited because of the limited grate area.
- The impurities settle at the bottom and thus prevents the water from heating up.
- The boiler tube should be kept small to reduce height. As a result, most of the available heat is lost through the chimney, as the tubes have very little time to heat.

Boiler Mountings:

The devices which are necessary for the safe operation of the boiler are called as boiler mountings. They are directly mounted on a boiler.

1) **Water level indicator:** It indicates the level of water in the boiler drum. Two water level indicators are fitted at the front end of the boiler drum.

2) **Pressure gauge:** Indicates the pressure of the steam in the boiler. Pressure Gauge usually mounted in the front end, at the top of the boiler

3) **Safety Valves:** When the pressure inside the boiler drum exceeds the desired level, the safety valves blows off the excess steam from the boiler. Safety valves are placed directly on the top of the boiler shell.

4) **Steam stop valve:** It regulates the flow of steam from the boiler. This valve is connected to a steam pipe for the flow of steam from the boiler to engine.

5) **Feed check valve:** It checks the level of water in the boiler drum. It is fitted over the shell below the normal water level of the boiler.

6) **Blow off valve:** Its function is to remove periodically the sediments and impurities collected at the bottom of the boiler.

7) **Fusible plug:** It is a device used to extinguish the fire in the furnace. It is fitted over the combustion chamber at appropriate place.

Boiler Accessories:

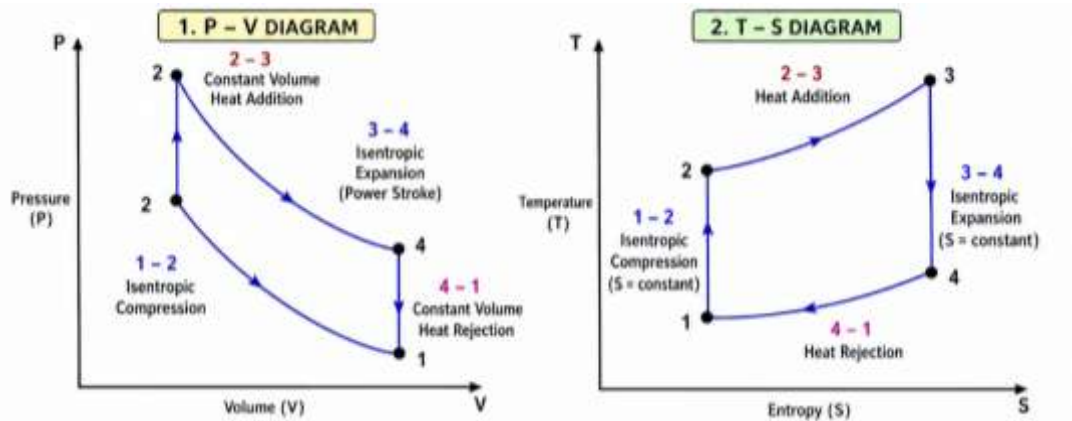
1. **Economizer:** The function of an economizer is to heat the feed water, before being supplied to the boiler, using the products of combustion (flue gas) discharged from the boiler. It is placed in between the boiler exit and the chimney entrance.
2. **Air preheater:** The function of an air preheater is to preheat the air being supplied to the furnace for combustion. It is placed between the economizer and the chimney entrance.
3. **A super heater:** is a device used to convert saturated steam or wet steam into dry steam. It is placed in the combustion chamber below the boiler drum.
4. **Feed pump:** pumps the water into the boiler at high pressure. It is usually fitted outside the boiler.
5. **Steam separator:** it is used to separate the water particles are present in the steam before enters the turbine or engines. It is placed closer to the engines.
6. **Steam trap:** it is used to drain off the condensed water accumulated in the steam pipes and steam separator without allowing the escape of high pressure steam from it.

Difference Between Water Tube & Fire tube boilers

Sl no	WATER TUBE BOILERS	FIRE TUBE BOILERS
1.	In water tube boilers, water is circulated inside the tubes and hot gases surround the water tubes	In fire tube boilers, hot gases pass inside the tube and water surrounds the tubes.
2.	Furnace is situated outside the boilers shell.	Furnace is situated inside the boiler shell
3.	Water circulates between the drum and the tubes maintaining a closed circuit.	Water circulates within the boilers drum only.
4.	Combustion space is large, complete combustion of fuel is possible	Combustion efficiency is low. Combustion takes place in a small space within the boiler shell.
5.	Steam generation rate is fast.	Steam generation rate is slow
6.	Evaporating capacity is high	Rate of evaporation is low
7.	Thermal efficiency is high	Thermal efficiency is low
8.	All parts of water tubes boilers are easily accessible for cleaning, inspection and repair	Cleaning, inspection and repairing is difficult due to inaccessible parts.
9.	High cost	Low cost
10.	Used in power plants	Used in process industries.

OTTO CYCLE

Otto cycle is an ideal air-standard cycle consisting of two isentropic processes and two constant volume processes, used to describe the working of spark ignition engines, where heat addition occurs at constant volume and converts thermal energy into mechanical work.



Process (1-2) – Adiabatic Compression : In the Otto cycle, adiabatic compression (process 1–2) is the stage where the air–fuel mixture is compressed by the piston moving from bottom dead center to top dead center without any heat transfer to or from the surroundings ($Q = 0$). As the volume decreases, the pressure and temperature of the mixture increase significantly due to the work done on the gas, making it ready for efficient combustion in the next process; this compression is also isentropic, meaning the entropy remains constant throughout the process.

Process (2-3) - Constant Volume Heat Addition : In the Otto cycle, constant volume heat addition (process 2–3) is the stage where the compressed air–fuel mixture undergoes combustion, and heat is added while the piston remains momentarily at top dead center, so the volume stays constant. During this process, the temperature and pressure of the gas increase rapidly due to the release of heat energy from fuel combustion, preparing the system for the expansion (power) stroke that follows.

Process (3-4) – Adiabatic Expansion: In the Otto cycle, adiabatic expansion (process 3–4) is the stage where the high-pressure, high-temperature gases produced after combustion expand as the piston moves from top dead center to bottom dead center without any heat transfer ($Q = 0$). During this process, the gas does work on the piston, causing the volume to increase while the pressure and temperature decrease, and this expansion is isentropic (constant entropy), making it the power-producing stroke of the cycle.

Process (4-1) – Constant Volume Heat Rejection: In the Otto cycle, constant volume heat rejection (process 4–1) is the stage where the burnt gases release heat to the surroundings while the piston is at bottom dead center, so the volume remains constant. During this process, the temperature and pressure of the gas decrease significantly as heat is rejected, bringing the system back to its initial state and completing the cycle.

Advantages of Otto Cycle

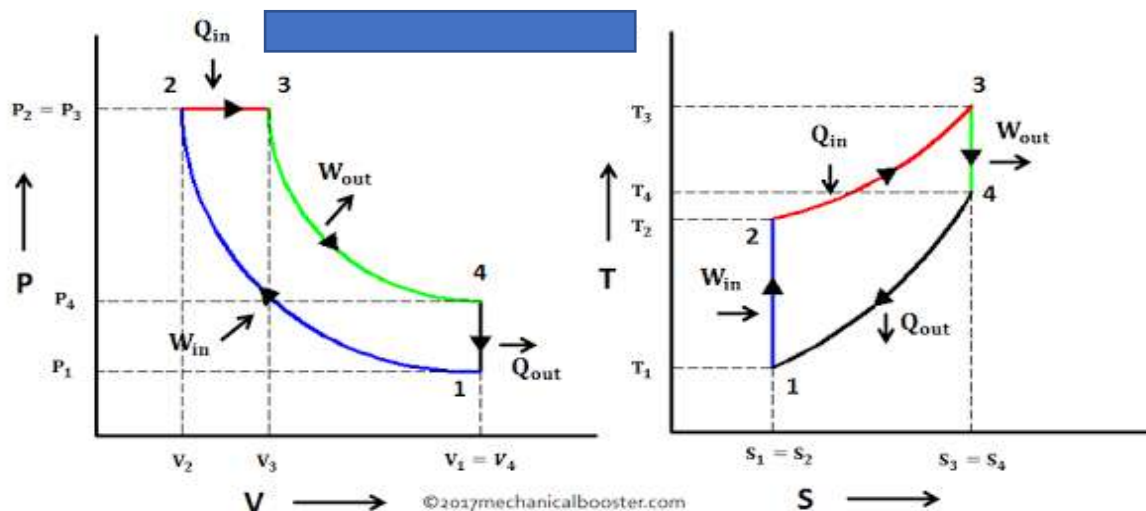
1. **Higher thermal efficiency** at high compression ratios.
2. Simple and compact engine design.
3. Smooth and quiet operation.
4. High engine speed capability.
5. Lower weight compared to diesel engines.
6. Lower initial cost.

Disadvantages of Otto Cycle

1. Lower efficiency than diesel cycle in practical operation.
2. Uses petrol, which is more expensive than diesel.
3. Compression ratio is limited due to knocking.
4. Higher fuel consumption.
5. Produces more carbon monoxide emissions.

DIESEL CYCLE

The Diesel cycle is an ideal thermodynamic cycle that represents the working of a diesel engine, in which air is compressed adiabatically, fuel is injected and burns at constant pressure, followed by expansion and heat rejection to complete the cycle.



Process (1-2) Adiabatic Compression: In the Diesel cycle, adiabatic compression (process 1–2) is the first stage where only air is compressed by the piston moving from bottom dead center to top dead center without any heat transfer ($Q = 0$). As the volume decreases, the pressure and temperature of the air increase significantly due to the work done on it, making the temperature high enough to ignite the fuel when it is injected in the next process, and this compression is also isentropic, meaning entropy remains constant.

Process (2-3) Constant Pressure Heat Addition: In the Diesel cycle, constant pressure heat addition (process 2–3) is the stage where fuel is injected into the highly compressed hot air and combustion takes place while maintaining nearly constant pressure. During this process, the volume increases as the piston starts moving downward, and the temperature of the gas rises due to heat release, preparing the system for the expansion (power) stroke.

Process (3-4) Adiabatic Expansion: In the Diesel cycle, adiabatic expansion (process 3–4) is the stage where the high-pressure, high-temperature gases expand as the piston moves from top dead center to bottom dead center without any heat transfer ($Q = 0$). During this process, the gas does work on the piston, causing the volume to increase while the pressure and temperature decrease, and this expansion is isentropic, making it the power-producing stroke of the cycle.

Process (4-1) Constant Volume Heat Rejection: In the Diesel cycle, constant volume heat rejection (process 4–1) is the stage where the burnt gases release heat to the surroundings while the piston is at bottom dead center, so the volume remains constant. During this process, the temperature and pressure of the gas decrease significantly as heat is rejected, bringing the system back to its initial state and completing the cycle.

Advantages of Diesel Cycle

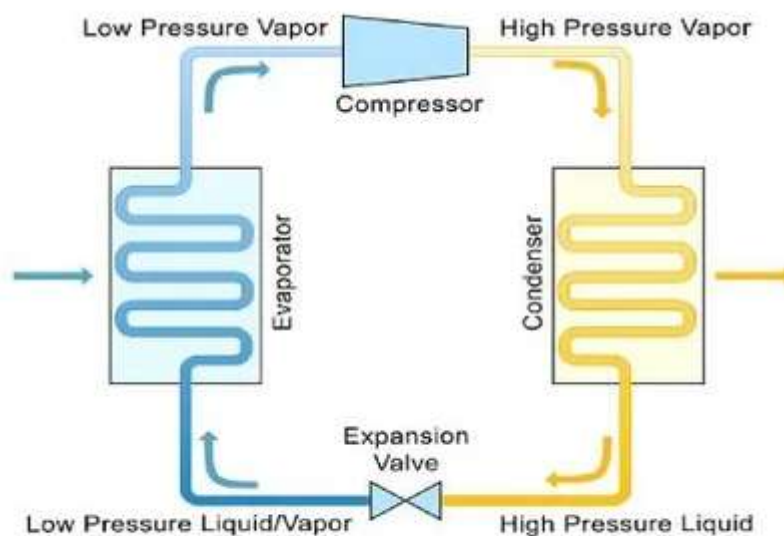
1. Higher thermal efficiency than Otto cycle.
2. Better fuel economy.
3. High torque output.
4. Suitable for heavy-duty applications.

Disadvantages of Diesel Cycle

1. Higher initial cost.
2. Engine is heavier and bulkier.
3. More noise and vibration.
4. Maintenance cost is high.

REFRIGERATION CYCLE

A vapour Compression Refrigeration is an improved air refrigeration system in which a suitable working substance is used as a refrigerant. This refrigerant evaporates and condenses at atmospheric pressure and temperature. The refrigerants usually used for this purpose are freons, Ammonia, Carbon dioxide and Sulphur-dioxide. Vapour Compression refrigeration systems are being used for all purposes of refrigeration varying from small domestic refrigerators to big air conditioning plants.



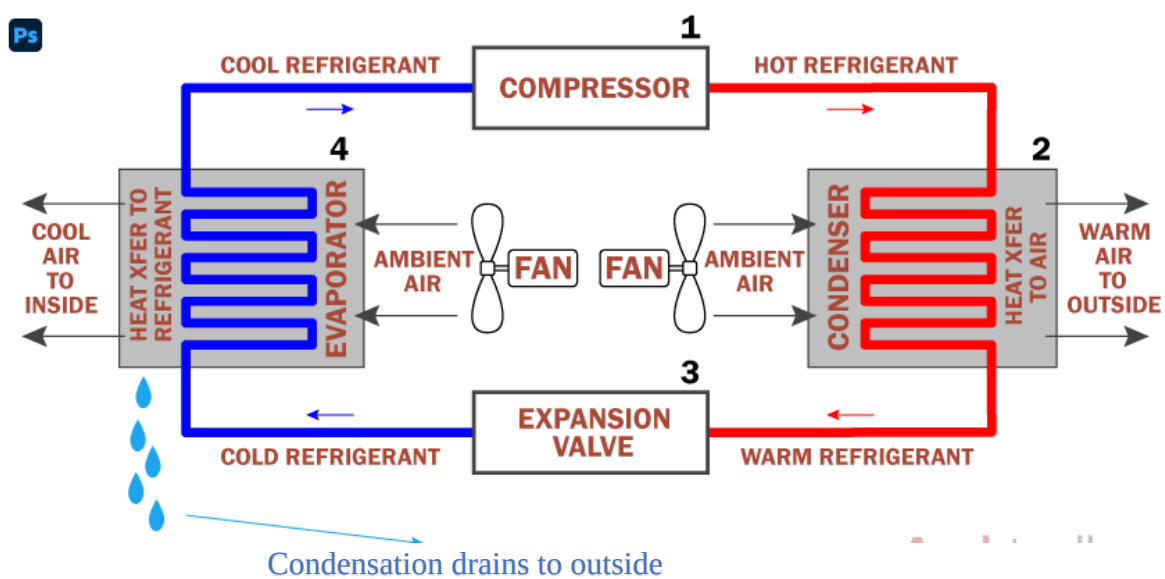
Compressor: The low-pressure and low-temperature vapour refrigerant from evaporator sucks into the compressor through suction value. In compressor it gets compressed to high pressure and elevated temperature. This high pressure and elevated temperature vapour refrigerant is discharged to condenser through outlet value of compressor.

Condenser: In condenser the high pressure and elevated temperature vapor refrigerant changes to high pressure and elevated temperature liquid refrigerant by rejecting latent heat in it (refrigerant phase change).

Expansion Valve: It is also called throttle valve. In expansion valve, the high pressure and elevated temperature liquid refrigerant changes to low pressure and low temperature by allowing to it pass at controlled rate.

Evaporator: An evaporator is a space where the products to be cooled are kept. It consists of cooling coils in which the refrigerant flows. In evaporator low pressure and low temperature liquid refrigerant becomes low pressure and low temperature vapour refrigerant by absorbing heat from the products (refrigerant phase change) and then refrigerant leaves to the compressor for next cycle of operation

Air Conditioning Cycle



Air Conditioning System Operation: The functioning of an air conditioning system follows a specific sequence:

- 1. Compression:** The refrigerant initially enters the compressor as a low-pressure superheated gas.
- 2. Compressor:** The compressor plays a crucial role by compressing the gas, transforming it into a high-pressure superheated gas.
- 3. Condensation:** Within the condenser, the gas undergoes cooling and a change of state into a vapour. Further cooling inside the condenser leads to the condensation of the refrigerant vapour, turning it into a high-pressure subcooled liquid.

4. Expansion: As the high-pressure liquid refrigerant passes through the metering device, it enters a low-pressure environment. This transition causes it to rapidly change into a vapour state.

5. Evaporation: The refrigerant vapour then enters the evaporator, where it absorbs heat from the area being cooled. This heat absorption causes the refrigerant to boil. While passing through the evaporator coil, the vapour becomes superheated, transitioning it into a gas state before it re-enters the compressor, restarting the cycle

Advantages of Air Conditioning Cycle

1. Provides comfortable indoor temperature.
2. Removes excess humidity from air.
3. Improves air quality and ventilation.
4. Helps in better sleep and productivity.

Disadvantages of Air Conditioning Cycle

1. High electricity consumption.
2. Installation cost is expensive.
3. Requires regular maintenance.

Applications of Air Conditioning Systems: Air conditioning systems find a multitude of applications:

- 1) In everyday life, they are utilized in homes, offices, cars, shopping malls, theatres and various other settings.
- 2) For residential spaces and small offices, Split Air Conditioners and Window AC units are commonly employed.
- 3) In larger commercial establishments and complexes, Ductable Air Conditioners, Variable Refrigerant Volume (VRV) systems and similar solutions are preferred.
- 4) For industrial purposes, especially in large-scale manufacturing or mass production facilities, Central Air Conditioning Systems, which may include Chillers, Heat Pumps and Air Handling Units (AHUs), are extensively used to maintain precise temperature control.

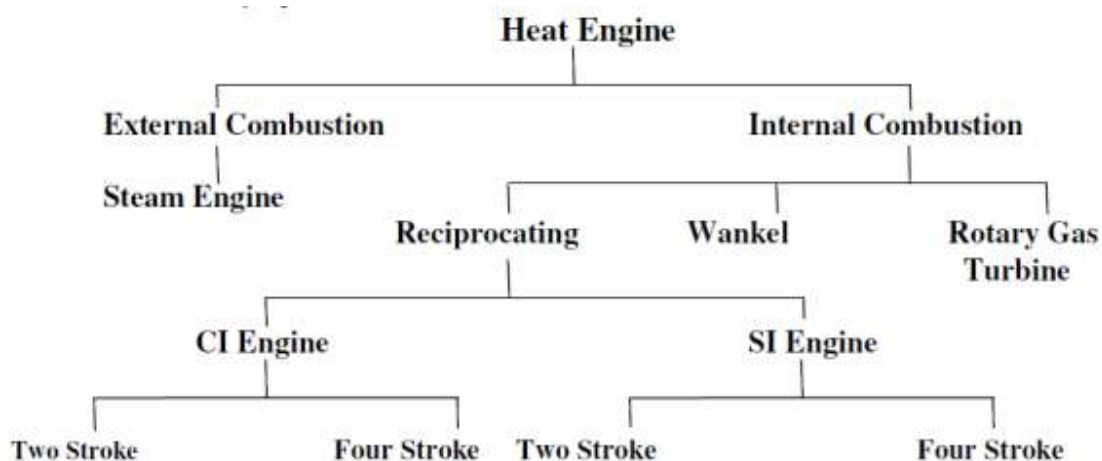
IC ENGINES

An Internal Combustion Engine (I.C. Engine) is a heat engine which converts chemical energy in fuel into heat or thermal energy and then to mechanical energy. To convert chemical energy into heat energy, fuel is to be burnt, that is, it must be subjected to combustion. If this combustion takes place inside the cylinder of a heat engine, then it is called an I.C. engine. On the other hand, if combustion takes place outside the engine (say in a boiler) and the working fluid is conveyed into a cylinder then it is called an external combustion engine.

HEAT ENGINES:

A heat engine is a machine, which converts heat energy into mechanical energy. The combustion of fuel such as coal, petrol, diesel generates heat. This heat is supplied to a working substance at high temperature. By the expansion of this substance in suitable machines, heat energy is converted into useful work. Heat engines can be further divided into two types:

- (i) External combustion and
- (ii) Internal combustion.



External combustion engine: Here, the working medium, the steam, is generated in a boiler, located outside the engine and allowed in to the cylinder to operate the piston to do mechanical work.

Internal combustion engine: In internal combustion engine, the combustion of fuel takes place inside the engine cylinder and heat is generated within the cylinder. This heat is added to the air inside the cylinder and thus the pressure of the air is increased tremendously. This high pressure air moves the piston which rotates the crank shaft and thus mechanical work is done. Most of the internal combustion engines are reciprocating engines with a piston that reciprocate back and forth in the cylinder.

Classification of Internal Combustion Engines:

Based on fuel used:

1. Diesel engine
2. Petrol engine
3. Gas engine

Diesel engine – Diesel is used as fuel

Petrol engine – Petrol is used as fuel

Gas engines – propane, butane or methane gases are used

Based ignition of fuel

1. Spark ignition engine (Carburetor type engines)
2. Compression ignition engine (injector type engines)

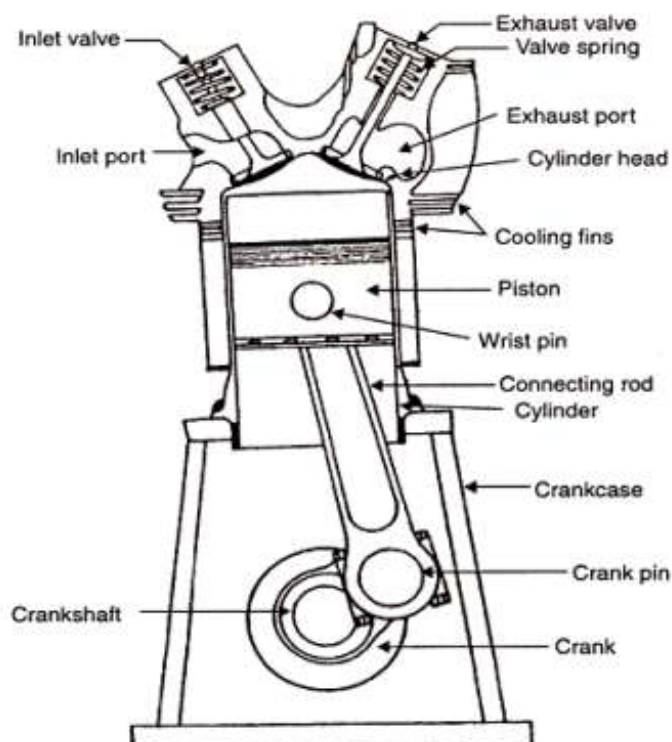
Spark ignition engine – A mixture of air and fuel is drawn in to the engine cylinder. Ignition of fuel is done by using a spark plug. The spark plug produces a spark and ignites the air- fuel mixture. Such combustion is called constant volume combustion (C.V.C.).

Compression ignition engine – In compression ignition engines air is compressed in to the engine cylinder,. Due to this the temperature of the compressed air rises to 700-900C. At this stage diesel is sprayed in to the cylinder in fine particles. Due to a very high temperature, the fuel gets ignited. This type of combustion is called constant pressure combustion (CP.C.) because the pressure inside the cylinder is almost constant when combustion is taking place.

Based on working cycle

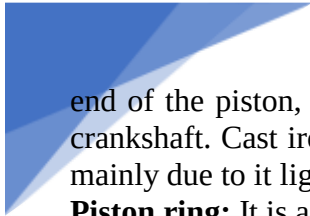
1. **Four stroke cycle engine** - When the cycle is completed in two revolutions of the crankshaft, it is called four stroke cycle engine.
2. **Two stroke cycle engine.** - When the cycle is completed in one revolution of the crankshaft, it is called two stroke cycle engine.

ENGINE COMPONENTS: Internal combustion engine consists of a number of parts which are given below :



Cylinder: It is a part of the engine which confines the expanding gases and forms the combustion space. It is the basic part of the engine. It provides space in which piston operates to suck the air or air-fuel mixture. The piston compresses the charge and the gas is allowed to expand in the cylinder, transmitting power for useful work. Cylinders are usually made of high grade cast iron.

Piston: It is a cylindrical part closed at one end which maintains a close sliding fit in the engine cylinder. It is connected to the connecting rod by a piston pin. The force of the expanding gases against the closed



end of the piston, forces the piston down in the cylinder. This causes the connecting rod to rotate the crankshaft. Cast iron is chosen due to its high compressive strength. Aluminum and its alloys preferred mainly due to its lightness.

Piston ring: It is a split expansion ring, placed in the groove of the piston. They are usually made of cast iron or pressed steel alloy. The function of the ring are as follows :

- a) It forms a gas tight combustion chamber for all positions of piston.
- b) It reduces contact area between cylinder wall and piston wall preventing friction losses and excessive wear.
- c) It controls the cylinder lubrication.
- d) It transmits the heat away from the piston to the cylinder walls. Piston rings are of two types:

(1) Compression ring and (2) Oil ring

Piston Pin: It is also called wrist pin or gudgeon pin. Piston pin is used to join the connecting rod to the piston.

Connecting rod: It is special type of rod, one end of which is attached to the piston and the other end to the crankshaft. It transmits the power of combustion to the crankshaft and makes it rotate continuously. It is usually made of drop forged steel.

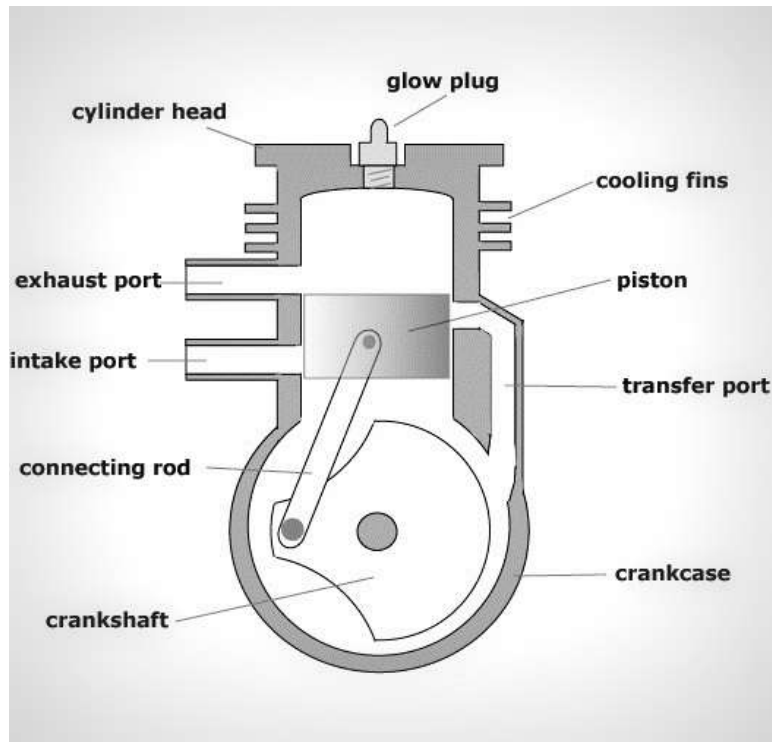
Crankshaft: It is the main shaft of an engine which converts the reciprocating motion of the piston into rotary motion of the flywheel. Usually the crankshaft is made of drop forged steel or cast steel. The space that supports the crankshaft in the cylinder block is called main journal, whereas the part to which connecting rod is attached is known as crank journal. Crankshaft is provided with counter weights throughout its length to have counter balance of the unit.

Crankcase: The crankcase is that part of the engine which supports and encloses the crankshaft and camshaft. It provides a reservoir for the lubricating oil. It also serves as a mounting unit for such accessories as the oil pump, oil filter, starting motor and ignition components. The upper portion of the crankcase is usually integral with cylinder block. The lower part of the crankcase is commonly called oil pan and is usually made of cast iron or cast aluminum. Crankshaft gear and it has twice as many teeth as that of the camshaft gear. For this reason, this gear is commonly called half time gear. Timing gear controls the timing of ignition, timing of opening and closing of valve as well as fuel injection timing.

Inlet Valve: It is that part of the engine through which air or air-fuel mixture enters into the engine cylinder. It is fitted by the side of the cylinder head.

Exhaust Valve: It is that part of the engine through which exhaust gases go out of the engine cylinder. It is capable of withstanding high temperature of burnt gases. It is fitted by the side of the cylinder head.

TWO STROKE CYCLE ENGINE (PETROL ENGINE): In two stroke cycle engines, the whole sequence of events i.e., suction, compression, power and exhaust are completed in two strokes of the piston i.e. one revolution of the crankshaft. There is no valve in this type of engine. Gas movement takes place through holes called ports in the cylinder. The crankcase of the engine is air tight in which the crankshaft rotates.

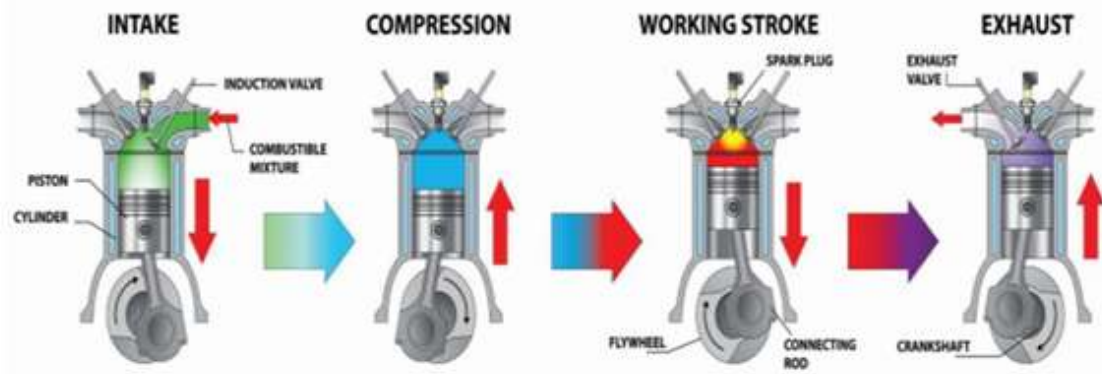


Upward stroke of the piston (Suction + Compression): When the piston moves upward it covers two of the ports, the exhaust port and transfer port, which are normally almost opposite to each other. This traps the charge of air- fuel mixture drawn already in to the cylinder. Further upward movement of the piston compresses the charge and also uncovers the suction port. Now fresh mixture is drawn through this port into the crankcase. Just before the end of this stroke, the mixture in the cylinder is ignited by a spark plug. Thus, during this stroke both suction and compression events are completed.

Downward stroke (Power + Exhaust): Burning of the fuel rises the temperature and pressure of the gases which forces the piston to move down the cylinder. When the piston moves down, it closes the suction port, trapping the fresh charge drawn into the crankcase during the previous upward stroke. Further downward movement of the piston uncovers first the exhaust port and then the transfer port. Now fresh charge in the crankcase moves in to the cylinder through the transfer port driving out the burnt gases through the exhaust port. Special shaped piston crown deflect the incoming mixture up around the cylinder so that it can help in driving out the exhaust gases. During the downward stroke of the piston power and exhaust events are completed.

FOUR STROKE CYCLE ENGINE (DIESEL/ PETROL ENGINE): In four stroke cycle engines the four events namely suction, compression, power and exhaust take place inside the engine cylinder. The four events are completed in four strokes of the piston (two revolutions of the crank shaft). This engine has got valves for controlling the inlet of charge and outlet of exhaust gases. The opening and closing of the valve is controlled by cams, fitted on camshaft. The camshaft is driven by crankshaft with the help of suitable gears or chains. The camshaft runs at half the speed of the crankshaft. The events taking place in I.C. engine are as follows:

1. Suction stroke
2. Compression stroke
3. Power stroke
4. Exhaust stroke



1. Suction stroke: During suction stroke inlet valve opens and the piston moves downward. Only air or a mixture of air and fuel are drawn inside the cylinder. The exhaust valve remains in closed position during this stroke. The pressure in the engine cylinder is less than atmospheric pressure during this stroke.

2. Compression stroke: During this stroke the piston moves upward. Both valves are in closed position. The charge taken in the cylinder is compressed by the upward movement of piston. If only air is compressed, as in case of diesel engine, diesel is injected at the end of the compression stroke and ignition of fuel takes place due to high pressure and temperature of the compressed air. If a mixture of air and fuel is compressed in the cylinder, as in case of petrol engine, the mixture is ignited by a spark plug.

3. Power stroke: After ignition of fuel, tremendous amount of heat is generated, causing very high pressure in the cylinder which pushes the piston downward. The downward movement of the piston at this instant is called power stroke. The connecting rod transmits the power from piston to the crank shaft and crank shaft rotates. Mechanical work can be taped at the rotating crank shaft. Both valves remain closed during power stroke.

4. Exhaust stroke: During this stroke piston moves upward. Exhaust valve opens and exhaust gases go out through exhaust valves opening. All the burnt gases go out of the engine and the cylinder becomes ready to receive the fresh charge. During this stroke inlet valve remains closed. Thus it is found that out of four strokes, there is only one power stroke and three idle strokes in four stroke cycle engines. The power stroke supplies necessary momentum for useful work.

Difference Between 4 stroke and 2 stroke

Sl.No	Four Stroke Engine	Two Stroke Engine
1.	One working stroke for every two revolutions of the crankshaft.	One working stroke for each revolutions of the crankshaft.
2.	Turing moment on the crankshaft is not even due to one working stroke for every two revolution of the crankshaft. Hence heavy flywheel is required and engine runs unbalanced.	Turing moment on the crankshaft is more even due to one working stroke for each revolution of the crankshaft. Lighter flywheel is required and engine runs balanced.
3.	Complicated lubricating system.	Simple lubricating system.
4.	Engine design is complicated.	Engine design is simple.
5.	Less mechanical efficiency due to more friction on many parts.	More mechanical efficiency due to less friction on a few parts.
6.	More output due to full fresh charge intake and full burnt gases exhaust.	Less output due to mixing of fresh charge with the burnt gases.
7.	Less fuel consumption and full burning of fuel.	More fuel consumption and fresh charge is mixed with exhaust gases.
8.	Engine consists of inlet and exhaust valves.	Engine consists of inlet and exhaust ports.

SI & CI ENGINE

S. NO.	Spark Ignition Engines (SI)	Compression Ignition Engines (CI)
1	It draws air fuel mixture into the cylinder during suction stroke	It draws only air into the cylinder during suction stroke.
2	Petrol engines operate with low pressure and temperature	Diesel engines operate with high pressure and temperature
3.	Pressure ranges from 6 to 12 bar Temperature ranges from 250°C to 300°C	Pressure ranges from 35 to 40 bar Temperature ranges from 600oC to 700oC
4	It is fitted with carburettor and spark plugs	It is fitted with fuel injection pump and injectors
5	The burning of fuel takes place at constant volume	The burning of fuel takes place at constant pressure
6.	Ignition of air fuel mixture takes place by an electric spark produced by spark plug	Ignition of air fuel takes placed by a injection of fuel into the hot compressed air.
7	Petrol engines are quality governed engines. The speed of petrol engines are controlled by varying the quantity of air fuel mixture.	Diesel engines are quantity governed engines. The speed of diesel engines is controlled by varying quality of air fuel mixture. (rich or weak mixture)
8	Petrol engines are widely used in automobiles and aeroplanes etc.,	Diesel engines are widely used in heavy vehicles, such as buses, lorries, trucks etc.,

Advantages and Disadvantages Of Two-Stroke Cycle Over Four Stroke Cycle Engines:

Advantages:

- 1) The two-stroke cycle engine gives one working stroke for each revolution of the crankshaft. Hence theoretically the power developed for the same engine speed and cylinder volume is twice that of the four-stroke cycle engine, which gives only one working stroke for every two revolutions of the crankshaft. However, in practice, because of poor scavenging, only 50- 60% extra power is developed.
- 2) Due to one working stroke for each revolution of the crankshaft, the turning moment on the crankshaft is more uniform. Therefore, a two-stroke engine requires a lighter flywheel.
- 3) The two-stroke engine is simpler in construction. The design of its ports is much simpler and their maintenance easier than that of the valve mechanism.
- 4) The power required overcoming frictional resistance of the suction and exhaust strokes is saved, resulting in some economy of fuel.
- 5) Owing to the absence of the cam, camshaft, rockers, etc. of the valve mechanism, the mechanical efficiency is higher.
- 6) The two-stroke engine gives fewer oscillations.
- 7) For the same power, a two-stroke engine is more compact and requires less space than a four-stroke cycle engine. This makes it more suitable for use in small machines and motorcycles.
- 8) A two-stroke engine is lighter in weight for the same power and speed especially when the crankcase compression is used.
- 9) Due to its simpler design, it requires fewer spare parts. 10) A two-stroke cycle engine can be easily reversed if it is of the valve less type.

Disadvantages:

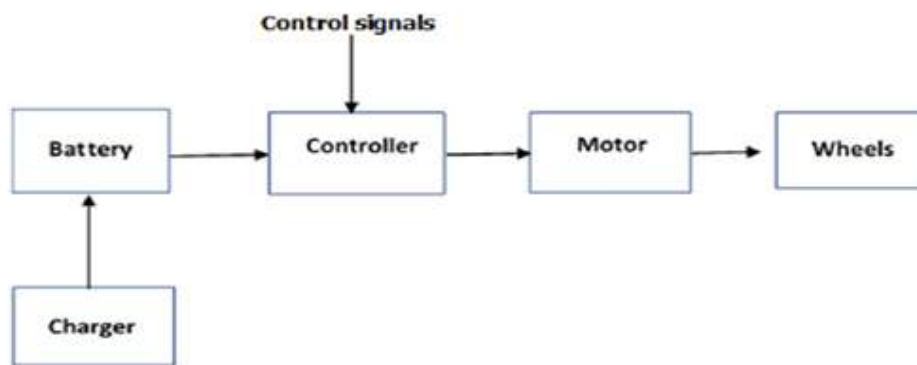
1. The scavenging being not very efficient in a two-stroke engine, the dilution of the charges takes place which results in poor thermal efficiency.
2. The two-stroke spark ignition engines do not have a separate lubrication system and normally, lubricating oil is mixed with the fuel. This is not as effective as the lubrication of a four-stroke engine. Therefore, the parts of the two-stroke engine are subjected to greater wear and tear.
3. In a spark ignition two-stroke engine, some of the fuel passes directly to the exhaust. Hence, the fuel consumption per horsepower is comparatively higher.
4. With heavy loads a two-stroke engine gets heated up due to the excessive heat produced. At the same time the running of the engine is not very smooth at light loads.
5. It consumes more lubricating oil because of the greater amount of heat generated. 6. Since the ports remain open during the upward stroke, the actual compression starts only after both the inlet and exhaust ports have been closed. Hence, the compression ratio of this engine is lower than that of a four-stroke engine of the same dimensions. As the efficiency of an engine is directly proportional to

its compression ratio, the efficiency of a two-stroke cycle engine is lower than that of a four-stroke cycle engine of the same size.

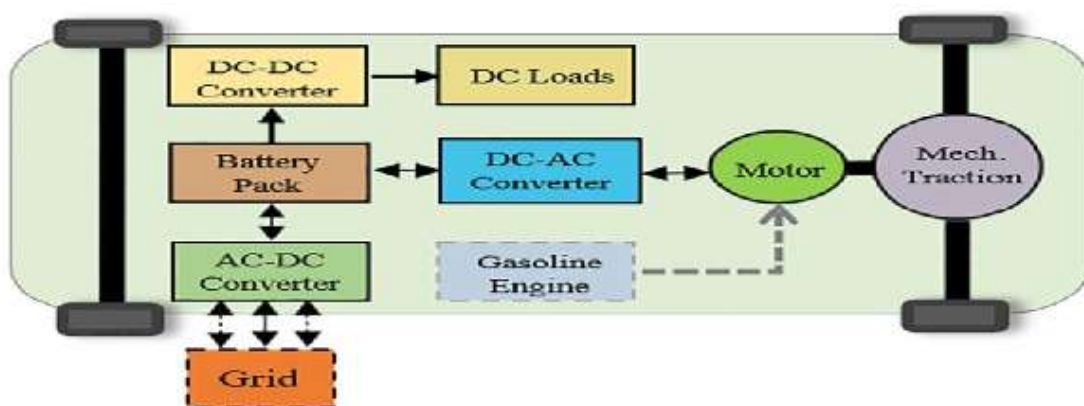
Electrical Vehicle:

The use of electric vehicles is an active action in conserving the environment. The use of these vehicles promotes the protection of the environment by reducing greenhouse gas emissions while promoting renewable energy sources that are less likely to amount to the carbon footprint.

An electric vehicle (EV) is a vehicle that uses one or more electric motors for propulsion. It can be powered by a collector system, with electricity from extravehicular sources, or it can be powered autonomously by a battery (sometimes charged by solar panels, or by converting fuel to electricity using fuel cells or a generator) EVs include, but are not limited to, road and rail vehicles, surface and underwater vessels, electric aircraft and electric spacecraft.



Block diagram of the electric vehicle



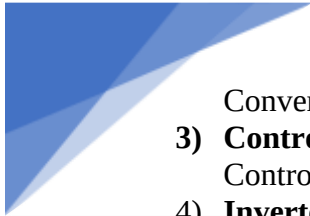
Structure of the electric vehicle

Components:

1) Battery Pack

Stores electrical energy required to run the vehicle. It is the main power source of an EV.

2) Electric Motor



Converts electrical energy from the battery into mechanical energy to drive the wheels.

3) Controller

Controls the speed and operation of the electric motor based on driver input.

4) Inverter

Converts DC power from the battery into AC power for the motor (in AC motor systems).

5) Charging Port

Used to connect the vehicle to an external power supply for charging.

6) On-board Charger

Converts AC electricity from the charging station into DC electricity to charge the battery.

7) Braking System

A braking system is a mechanism used to slow down or stop a vehicle by reducing the speed of the wheels.

Advantages:

1. Environmentally friendly

EVs produce no tailpipe emissions, which helps reduce air pollution and greenhouse gases. This is especially important for improving urban air quality and fighting climate change.

2. Lower running costs

Electricity is generally cheaper than fuel, and EVs are more energy-efficient. You spend less on charging than on petrol or diesel.

3. Less maintenance

EVs have fewer moving parts (no engine oil, fewer fluids, no exhaust system), so they require less maintenance and have lower servicing costs.

4. Quiet operation

Electric motors are much quieter than internal combustion engines, reducing noise pollution.

5. High efficiency

EVs convert a higher percentage of energy from the battery into motion, making them more efficient than conventional vehicles.

6. Instant torque and smooth driving

Electric motors deliver power instantly, resulting in quick acceleration and a smooth driving experience.

7. Government incentives

Many governments offer tax credits, subsidies, or reduced registration fees to encourage EV adoption.

8. Renewable energy compatibility

EVs can be charged using renewable energy sources like solar or wind, making them even more sustainable.

Disadvantages:

1) High initial cost

Electric vehicles are usually more expensive to buy than petrol or diesel vehicles because of the battery cost.

2) Limited driving range

Many EVs can travel only a limited distance on a full charge compared to fuel vehicles.

3) Charging takes time

Charging an EV takes much longer than filling petrol or diesel in a vehicle.

4) Charging station availability

Charging infrastructure is still limited in many areas, especially rural locations.

5) Battery replacement cost

EV batteries are expensive to replace after their lifespan decreases.

6) Reduced battery performance

Battery efficiency may reduce in very hot or very cold weather conditions.

7) Electricity dependency

EVs depend completely on electricity, so charging can be difficult during power cuts.

8) Limited heavy-load performance

Some EVs may not perform efficiently for long-distance heavy-load transportation.

9) Battery disposal issues

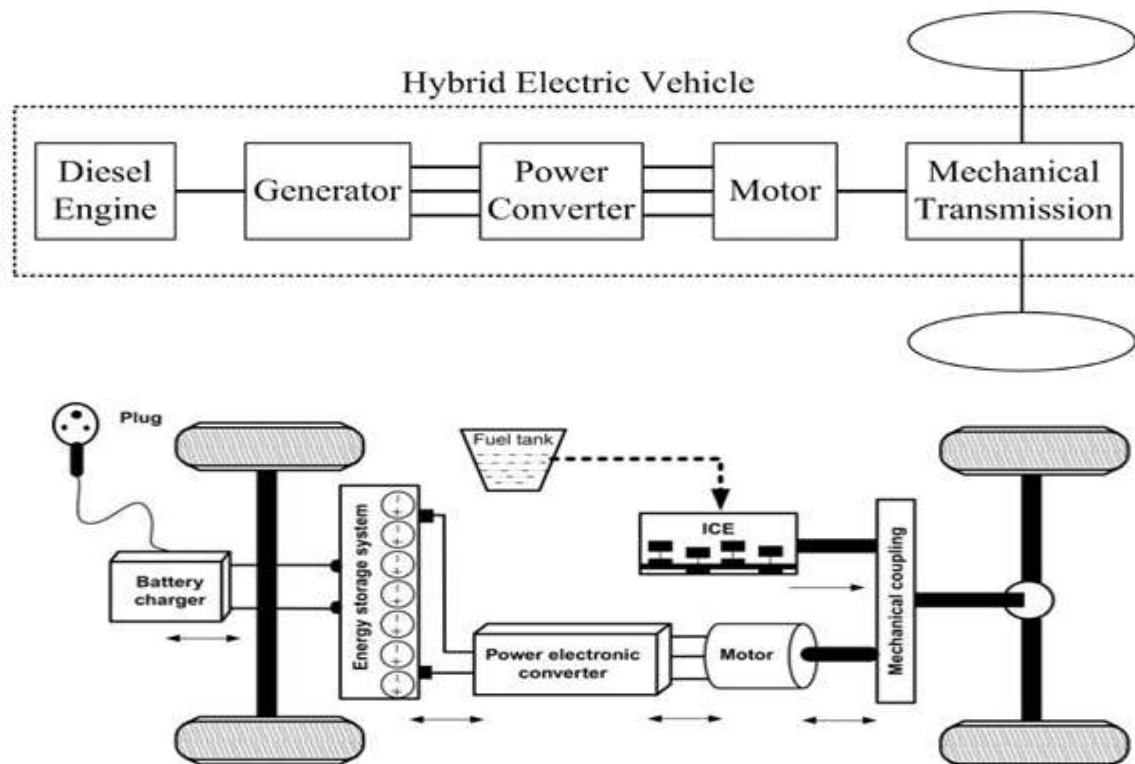
Used batteries require proper recycling and disposal to avoid environmental problems.

10) Range anxiety

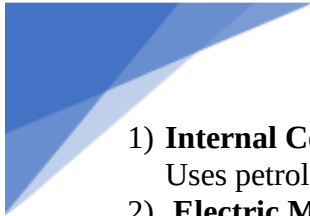
Drivers may worry about the battery getting discharged before reaching a charging station.

Hybrid Vehicles:

When compared to conventional vehicles, hybrids offer better power and fuel efficiency as they combine the benefits of high fuel efficiency and low emissions. When hybrid vehicles are cruising or while braking, the result is excess power which is used to charge the batteries. Hybrid electric vehicles are powered by an internal combustion engine and an electric motor, which uses energy stored in batteries. A hybrid electric vehicle cannot be plugged in to charge the battery. Instead, the battery is charged through regenerative braking and by the internal combustion engine.



Components of Hybrid Vehicle:

- 
- 1) **Internal Combustion Engine (IC Engine)**
Uses petrol or diesel to produce power for driving the vehicle.
 - 2) **Electric Motor**
Assists the engine or independently drives the vehicle using electrical energy.
 - 3) **Battery Pack**
Stores electrical energy required for the electric motor.
 - 4) **Fuel Tank**
Stores fuel for the IC engine.
 - 5) **Generator**
Converts mechanical energy from the engine into electrical energy to charge the battery.
 - 6) **Power Control Unit (PCU)**
Controls the flow of electrical power between the battery and motor.
 - 7) **Transmission System**
Transfers power from the engine and motor to the wheels.
 - 8) **Inverter / Converter**
Converts DC power from the battery into AC power for the motor and vice versa.
 - 9) **Regenerative Braking System**
Recovers energy during braking and stores it in the battery.
 - 10) **Charging System**
Charges the battery using the engine or external power source (in plug-in hybrids).
 - 11) **Vehicle Control Unit (VCU)**
Controls and coordinates the operation of engine, motor, and battery systems.
 - 12) **Cooling System**
Maintains safe temperature of the engine, motor, and battery.

Advantages

1. Better Fuel Efficiency

Hybrid vehicles use both a fuel engine and an electric motor, so fuel consumption is lower compared to normal petrol or diesel vehicles.

2. Less Pollution

Hybrids produce lower harmful gas emissions, which helps reduce air pollution and protects the environment.

3. Regenerative Braking

During braking, energy is converted into electricity and stored in the battery. This improves efficiency.

4. Smooth and Quiet Operation

Electric motors run quietly and provide smooth driving, especially at low speeds.

5. Reduced Fuel Cost

Because fuel usage is less, running cost becomes lower over time.

6. Automatic Switching

The vehicle automatically changes between engine power and electric power depending on driving conditions.

7. Better Mileage in City Traffic

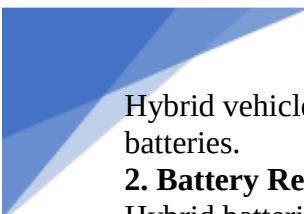
Hybrid vehicles perform well in stop-and-go traffic because the electric motor works efficiently at low speeds.

8. Lower Engine Wear

Since the electric motor supports the engine, mechanical stress on the engine is reduced.

Disadvantages

1. High Initial Cost



Hybrid vehicles are more expensive than conventional vehicles due to advanced technology and batteries.

2. Battery Replacement Cost

Hybrid batteries are costly to replace after their lifespan ends.

3. Complex System

The combination of engine, motor, and battery makes the system more complicated.

4. Higher Maintenance Difficulty

Repairs and servicing require trained technicians and special equipment.

5. Heavy Weight

Hybrid vehicles are heavier because of additional batteries and electric components.

6. Limited Electric Range

Most hybrid vehicles cannot travel long distances using only electric power.

7. Less Boot Space

Battery placement may reduce luggage/storage space.

8. Safety Concerns

High-voltage electrical systems require extra safety precautions during maintenance.